Lecture 23: The Dot Product

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Define and compute the dot product of two vectors in both two and three dimensions.
- ☐ Determine the angle between two vectors using the dot product.
- ☐ Verify whether two vectors are orthogonal.
- ☐ Apply properties of the dot product in vector algebra.
- ☐ Apply the dot product to solve problems involving projections of one vector onto another.

Angle Between Vectors

When two nonzero vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ are placed so their initial points coincide, they form an angle θ of measure $0 \leq \theta \leq \pi$.

Angle Between Two Vectors

The angle θ between two nonzero vectors $\boldsymbol{u} = \langle u_1, u_2, u_3 \rangle$ and $\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle$ is given by

$$\theta = \cos^{-1}\left(\frac{u_1 v_1 + u_2 v_2 + u_3 v_3}{|\boldsymbol{u}||\boldsymbol{v}|}\right).$$

Dot Product

Given two vectors $\boldsymbol{u} = \langle u_1, u_2, u_3 \rangle$ and $\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle$, the dot product $(\boldsymbol{u} \cdot \boldsymbol{v})$ is the scalar

$$\boldsymbol{u} \cdot \boldsymbol{v} = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

Example:

(a) $\langle 1, -2, -1 \rangle \cdot \langle -6, 2, -3 \rangle$

(b) $\left(\frac{1}{2}\boldsymbol{i} + 3\boldsymbol{j} + \boldsymbol{k}\right) \cdot (4\boldsymbol{i} - \boldsymbol{j} + 2\boldsymbol{k})$

Dot Product

Dot Product and Angle

The angle of two nonzero vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ is $\theta = \cos^{-1}\left(\frac{\boldsymbol{u}\cdot\boldsymbol{v}}{|\boldsymbol{u}||\boldsymbol{v}|}\right)$.

The dot product of two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ is given by $\boldsymbol{u}\cdot\boldsymbol{v} = |\boldsymbol{u}||\boldsymbol{v}|\cos\theta$.

Example: Find the angle between $\boldsymbol{u} = \boldsymbol{i} - 2\boldsymbol{j} - 2\boldsymbol{k}$ and $\boldsymbol{v} = 6\boldsymbol{i} + 3\boldsymbol{j} + 2\boldsymbol{k}$.

Orthogonal Vectors

Two non-zero vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ are orthogonal (or perpendicular) if $\boldsymbol{u} \cdot \boldsymbol{v} = 0$.

Example: Determine if the following vectors are orthogonal.

(a) $\boldsymbol{u} = \langle 3, -2 \rangle$, $\boldsymbol{v} = \langle 4, 6 \rangle$

(b) $\boldsymbol{u} = 3\boldsymbol{i} - 2\boldsymbol{j} + \boldsymbol{k}$, $\boldsymbol{v} = 2\boldsymbol{j} + 4\boldsymbol{k}$

(c) $\boldsymbol{0}$, $\boldsymbol{u} = \langle 1, 2, 3 \rangle$

Dot Product Properties

If $\boldsymbol{u}$, $\boldsymbol{v}$ and $\boldsymbol{w}$ are any vectors and c is a scalar, then

1. $\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{u}$
2. $(c\boldsymbol{u}) \cdot \boldsymbol{v} = \boldsymbol{u} \cdot (c\boldsymbol{v}) = c(\boldsymbol{u} \cdot \boldsymbol{v})$
3. $\boldsymbol{u} \cdot (\boldsymbol{v} + \boldsymbol{w}) = \boldsymbol{u} \cdot \boldsymbol{v} + \boldsymbol{u} \cdot \boldsymbol{w}$
4. $\boldsymbol{u} \cdot \boldsymbol{u} = |\boldsymbol{u}|^2$
5. $\boldsymbol{0} \cdot \boldsymbol{u} = 0$

Vector Projections

The vector projection of $\boldsymbol{u} = \overrightarrow{PQ}$ onto a nonzero vector $\boldsymbol{v} = \overrightarrow{PS}$ is the vector $\overrightarrow{PR}$ determined by dropping a perpendicular from Q to the line PS, denoted by $\mathrm{proj}_{\boldsymbol{v}}\boldsymbol{u}$ (read as the vector projection of $\boldsymbol{u}$ onto $\boldsymbol{v}$).

You may think this as the 'shadow' of $\boldsymbol{u}$ in the direction of $\boldsymbol{v}$. It is also sometimes referred as the component of $\boldsymbol{u}$ in the direction of $\boldsymbol{v}$.

$$\begin{aligned}
\mathrm{proj}_{\boldsymbol{v}}\boldsymbol{u} &= (|\boldsymbol{u}|\cos\theta)\frac{\boldsymbol{v}}{|\boldsymbol{v}|} \\
&= \left(\frac{\boldsymbol{u}\cdot\boldsymbol{v}}{|\boldsymbol{v}|}\right)\frac{\boldsymbol{v}}{|\boldsymbol{v}|} \\
&= \left(\frac{\boldsymbol{u}\cdot\boldsymbol{v}}{|\boldsymbol{v}|^2}\right)\boldsymbol{v}.
\end{aligned}$$

Vector Projections

Example: Find the vector projection of $\boldsymbol{u} = 6\boldsymbol{i} + 3\boldsymbol{j} + 2\boldsymbol{k}$ onto $\boldsymbol{v} = \boldsymbol{i} - 2\boldsymbol{j} - 2\boldsymbol{k}$ and the scalar component of $\boldsymbol{u}$ in the direction of $\boldsymbol{v}$.

Vector Projections

Example: Find the vector projection of $\boldsymbol{u} = 5\boldsymbol{i} + 2\boldsymbol{j}$ onto $\boldsymbol{v} = \boldsymbol{i} - 3\boldsymbol{j}$ and the scalar component of $\boldsymbol{u}$ in the direction of $\boldsymbol{v}$.

Vector Projections

Example: Show that the vector $\boldsymbol{u} - \text{proj}_{\boldsymbol{v}}\boldsymbol{u}$ is orthogonal to the projection vector $\text{proj}_{\boldsymbol{v}}\boldsymbol{u}$.